ANIL S A

AI ENGINEER

PORTFOLIO LINKEDIN

CONTACT

7411956534

anilangadi.26@gmail.com

Bengaluru,Karnataka,India

SKILLS

Programming

- Python, SQL,Java

Machine Learning

- Scikit-learn
- Feature Engineering
- Model Optimization
- Predictive Modeling,Model Monitoring
- Hyperparameter Tuning,Cross Validation

Deep Learning

- TensorFlow, PyTorch, HuggingFace, CNN
- LSTM, Transformers

Multimodal AI

- Speech-to-Text (Whisper), NLP, Image Classification

Generative AI

- LLMs, RAG Pipelines, Embeddings
- Agentic AI system (LangGraph / AutoGen)
- Vector Databases
- Prompt Engineering

Backend & APIs

- FastAPI, REST APIs
- Model Serving

MLOps & Deployment

- Docker
- Git
- Streamlit
- Model Deployment

Data & Databases

- PostgreSQL,Pandas,NumPy
- CI/CD Pipelines (GitHub Actions)
- Kubernetes (basic knowledge)

Cloud & Infrastructure

- AWS / Azure fundamentals

PROFILE SUMMARY

AI Engineer with 2+ years of experience building production-grade Machine Learning and Generative AI systems. Experienced in developing end-to-end ML pipelines, LLM-powered applications, and scalable APIs using Python, TensorFlow, PyTorch, and FastAPI. Skilled in Retrieval Augmented Generation (RAG), NLP, model optimization, and containerized deployment using Docker. Passionate about building enterprise AI systems integrating LLMs, vector databases, and real-time data processing.

EDUCATION

- **RNS INSTITUTE OF TECHNOLOGY**
  Master of Computer Applications
  8.0 CGPA
  2021-2023

- **RNS First Grade College**
  Bachelor of Computer Application
  8.26 CGPA
  2018-2021

WORK EXPERIENCE

AI Engineer
Cognizant Technology Solutions
Dec 2023 – Present

- Built end-to-end ML pipelines for enterprise analytics improving model accuracy by 12% and reducing inference latency by 30%. including data preprocessing, feature engineering, model training, evaluation, and deployment.
- Built regression and classification models using Random Forest, SVM, Logistic Regression, and SVR improving performance from R^2 0.78 to 0.87.
- Designed scalable NLP and transformer-based systems for enterprise AI assistants enabling semantic search and contextual query understanding.
- Implemented Retrieval-Augmented Generation (RAG) pipelines using embeddings and vector databases to improve information retrieval accuracy.
- Built deep learning models including CNNs and LSTMs for image classification and time-series forecasting applications.
- Developed REST APIs using FastAPI for real-time ML inference and integrated them with enterprise workflows.
- Containerized ML services using Docker, enabling scalable and reproducible deployment environments.
- Applied hyperparameter tuning and cross-validation techniques to optimize model performance and prevent overfitting.
- Designed embedding-based semantic search systems using cosine similarity for knowledge retrieval and recommendation tasks.
- Collaborated with cross-functional teams including backend developers, data engineers, and product stakeholders to integrate ML models into production systems and deliver AI-driven features.
- Explainable AI: Implemented model interpretability using LIME, LangChain, SHAP, and BertViz to explain predictions of ML, DL, and LLM models.

INTERNSHIP

Developed GenAI solutions using open-source LLMs including LLaMA and Stable Diffusion models.

• Implemented parameter-efficient fine-tuning (LoRA / QLoRA) reducing training compute by 70%.

• Built RAG pipelines with vector databases for document retrieval and contextual generation.

• Used Ray distributed computing for scalable model training and experimentation.

INTERNSHIP PROJECTS

Movie Recommendation using Machine Learning

Built a Movie Success Prediction model using Machine Learning, performing EDA, feature engineering, and regression modeling on movie metadata (genre, budget, ratings, cast) to identify key factors influencing box-office performance.

Car Price Prediction using Machine Learning

Built a Car Price Prediction model using Machine Learning, applying EDA, feature engineering, and regression algorithms to predict vehicle prices based on features like brand, year, mileage, fuel type, and transmission.

LANGUAGES

- English: Fluent
- Hindi: Fluent
- Kannada: Fluent

HOBBIES

- PC Gaming
- Reading
- Researching
- Problem-Solving
- Learning
- Exploring

CERTIFICATIONS

- Machine Learning (Udemy)
- Generative AI (Udemy)
- Python (Glovish Technology)
- Vibe Coding (Cognizant)

PROJECTS

OneC AI Assistant (Enterprise AI Platform)

- Built OneC AI Assistant, an enterprise AI platform enabling employees to perform HR and project operations (leave, timesheets, skill updates, project search) using natural language.
- Designed an LLM-powered enterprise assistant enabling employees to perform HR and project operations through natural language queries.
- Implemented intent classification and workflow automation to trigger backend actions via secure API integrations.
- Designed semantic search with embeddings and cosine similarity for project discovery and skill-to-project matching.
- Engineered role-based access control and secure data retrieval to protect sensitive enterprise information.
- Deployed scalable AI services using FastAPI, Docker, and vector databases, enabling low-latency enterprise AI responses.
- Tech Stack: Python, FastAPI, Whisper, HuggingFace, Vector DB, Docker.

AI Diagnostic System for Leukemia Detection (Vibro-Acoustic Sensor Data)

- Built an AI-based diagnostic system to detect leukemia patterns from vibro-acoustic and physiological sensor data (EEG, ECG, EMG) using advanced signal processing and ML models.
- Engineered time-domain and frequency-domain features from biomedical signals and applied supervised & unsupervised learning to improve model accuracy and generalization.
- Improved diagnostic performance AUC from 0.52 → 0.94 through feature optimization, model tuning, and robust evaluation.
- Developed automated markdown reports and visual analytics (graphs, bar plots) using multi-agent workflows to support scalable and privacy-conscious clinical analysis.
- Tech Stack: Python, Scikit-learn, Pandas, NumPy, Signal Processing

GenAI Powered Call Analytics Copilot

- Built a GenAI Call Analytics Copilot using Whisper + LLMs for real-time speech-to-text, summarization, and sentiment analysis, improving support efficiency by ~60%.
- Developed RAG-based semantic search with embeddings and vector databases to analyze live and historical call data for issue detection and insights.
- Implemented speaker diarization, time-segment sentiment analytics, and automated email summaries for agent assist and post-call intelligence.
- Deployed scalable services using FastAPI, and LLM pipelines for near real-time conversational analytics.
- Tech Stack:
- Python, Whisper, HuggingFace, FastAPI, Vector DB, Docker

Multimodal AI (Vision + LLM)

- Developed a Multimodal AI system integrating Computer Vision and Large Language Models to enable image understanding, visual question answering, and contextual reasoning.
- Implemented vision encoders with embedding pipelines and LLM inference to convert visual inputs into meaningful textual insights.
- Built scalable pipelines using Python, Vision Transformers (ViT), vector embeddings, and LLM APIs for multimodal interaction.
- Integrated RAG-based contextual retrieval to enhance response accuracy and reduce hallucinations in visual-text reasoning tasks.